Technical report - Apertus Difference Ranker

- **Track:** Track 2A - UZH: Detecting Cross-Lingual Semantic Differences in Swiss Government Websites
- **Event:** Online
- **Team:** Chenfei Wen (individual participant)
- **Demo:** Reproducible Docker result replay via make run

1. Summary

Related Swiss government pages can diverge because of translation mistakes or asynchronous updates. We built a token-level cross-lingual semantic-difference ranker around frozen Apertus v1.5 representations. The pipeline performs bidirectional whole-document DiffAlign, selects semantic-sensitive layers using development-training documents, and compares four lightweight supervised models. On the untouched 168-document test split, the best system-histogram gradient boosting over five layers of alignment features plus POS/NER-reaches **0.3275 macro Spearman**. Raw DiffAlign reaches **0.2632**, so the final model improves it by **0.0643 absolute** (24.4% relative). A complete, identically controlled 70B ablation improves raw DiffAlign to 0.2737 but reaches only 0.3168 with its best lightweight head; scaling therefore does not replace the more efficient 8B primary system.

2. Architecture

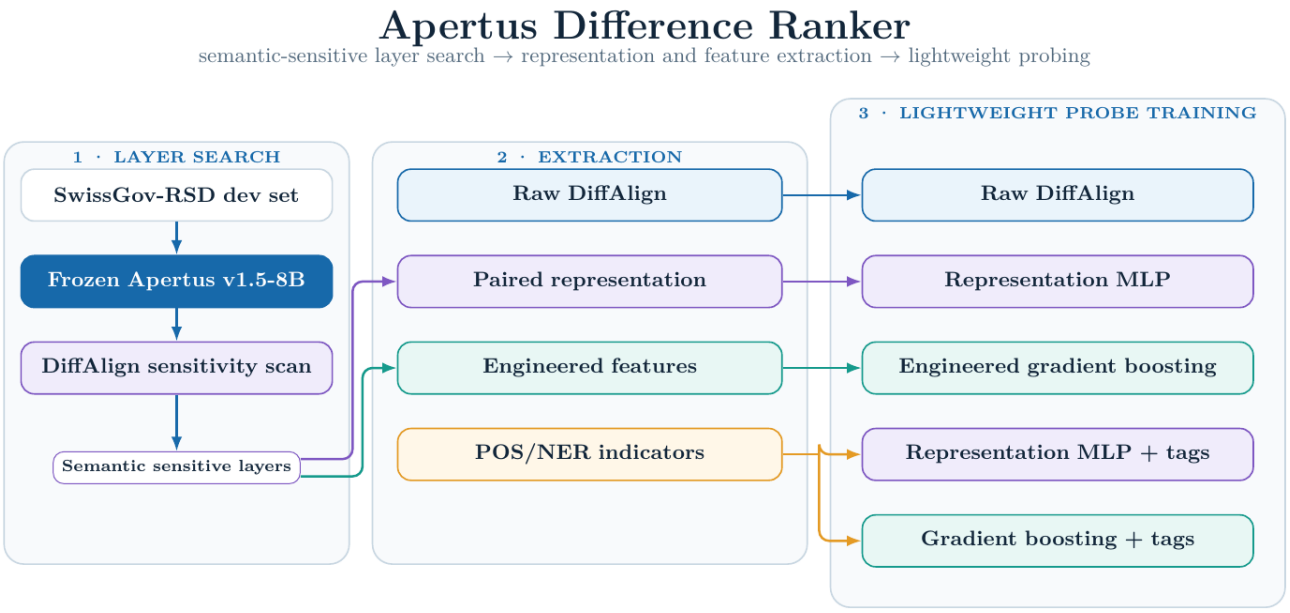


Figure 1. End-to-end training and evaluation paradigm. All layer selection, feature design, fitting, early stopping, and model choice use only the official development split; the test split is reserved for final evaluation.

The pipeline has five stages:

1. **Frozen extraction.** Each side of a bilingual document pair is encoded independently with Apertus. SwissGov whitespace words are mapped to one or more Apertus subwords and pooled back to the benchmark tokenization.
2. **Layer scan.** Evaluate DiffAlign at every layer on the 453-document fitting partition. Use the best layer for paired representations and the top five for engineered features, without using validation or test data.
3. **Feature construction.** The paired-state branch uses raw source and nearest target states. The engineered branch summarizes each token's cross-document cosine row and its local context. Optional POS/NER one-hot features are concatenated to either branch.
4. **Lightweight fitting.** Two representation MLPs and two histogram gradient boosters are trained while every Apertus parameter remains frozen.

5. **Frozen evaluation.** Selected layers, tag vocabularies, and fitted weights are applied to all 168 test documents without adaptation.

The data flow and leakage boundary are shown in docs/ARCHITECTURE.md.

2.1 Raw DiffAlign

For each source subword, DiffAlign finds the maximum cosine similarity to any subword in the counterpart document and converts it to a continuous difference score. Scores are averaged over the Apertus subwords belonging to each SwissGov word. The computation is repeated in both language directions.

2.2 Paired-representation MLP

At selected layer 14, each source word is paired with its cosine-nearest target word. For hidden size d=4096, the MLP input is:

```
[source_d, target_d, abs(source_d-target_d), source_d*target_d]
```

The untagged width is 4d = 16,384; the linguistic version appends 23 columns. The head is:

```
LayerNorm(input) -> Linear(input, 128) -> GELU -> Dropout(0.1)
-> Linear(128, 1) -> Sigmoid
```

The untagged head has 2,130,177 trainable parameters; Apertus has none. MLPs train for at most 20 epochs. Validation macro Spearman is checked after every epoch, the best state is retained, and training stops after five consecutive non-improving epochs.

2.3 Engineered gradient boosting

For each of selected layers 14-18, eight features are computed:

1. top-1 cosine distance;
2. mean top-5 cosine distance;
3. top-1/top-2 margin;
4. normalized alignment entropy;
5. cosine-row mean;
6. cosine-row standard deviation;
7. mutual-nearest-neighbour indicator;
8. source hidden-state norm.

Previous- and next-token top-1 distances add ten local-context values across five layers. Four static values encode subword count, relative position, log character length, and document side. The untagged vector therefore has 5x8 + 5x2 + 4 = 54 features.

The regressor is HistGradientBoostingRegressor with 300 iterations, learning rate 0.05, at most 31 leaf nodes, L2 regularization 1.0, and seed 42. It uses all 300 iterations; validation is diagnostic rather than an early-stopping signal.

2.4 Linguistic features

The tagged variants append 23 fixed-width one-hot columns:

- six NER categories: O, PER, ORG, LOC, MISC, PUNCT;
- seventeen Universal POS categories.

NER comes from Davlan/xlm-roberta-base-ner-hrl; POS comes from wietsedv/xlm-roberta-base-ft-udpos28-en. These auxiliary models do not supply semantic embeddings or alignment scores. They run as frozen taggers, are unloaded before Apertus extraction, and contribute only token-class labels. Punctuation is detected from observable token/POS information, never from test gold labels.

2.5 Objective and imbalance

Punctuation (gold = -1) is excluded. In MLP training, tokens with positive semantic difference receive total weight 5 versus weight 1 for zero-difference tokens. The loss is weighted Smooth-L1 plus a pairwise logistic ranking term with weight 0.2 and temperature 0.1. Gradient boosting uses the same sample weights.

3. Use of Apertus

- **Model:** swiss-ai/Apertus-v1.5-8B
- **Role:** frozen local representation extraction and all semantic alignment features
- **Execution:** local weights on CSCS Clariden GPU nodes
- **Selected layers:** representation layer 14; engineered layers 14-18
- **Test passes:** 336 for 168 documents, one pass per document side
- **Transformers implementation:** Swiss AI fork at commit 3797303dda74844e3d1f8977ff5518bb91f818b4

No non-Apertus embedding model contributes to the semantic comparison. The two auxiliary XLM-R models provide only coarse POS and NER categories as disclosed above. No Apertus parameter is updated.

For the controlled 70B ablation, Apertus also remains frozen. Development-only selection chooses representation layer 17 and engineered layers 17, 19, 18, 16, and 20. The architecture, training split, objectives, linguistic taggers, and test-isolation procedure are otherwise unchanged.

4. Data

We use ZurichNLP/SwissGov-RSD, a CC BY 4.0 benchmark of multilingual Swiss government document pairs with token-level semantic-difference labels. Hugging Face exposes 504 development pairs and 168 test pairs for the evaluated English-German, English-French, and English-Italian directions.

Using seed 42, the 504 development pairs are divided by document and language:

- 453 fitting documents;
- 51 validation documents, exactly 17 per language pair.

The test split is excluded from layer selection, fitting, early stopping, and feature design. Dataset records and model weights remain in their original repositories and are not committed. No additional personal data is collected.

5. Evaluation

The primary metric is token-level Spearman correlation for EN-DE, EN-FR, and EN-IT. We report each direction, their unweighted macro average, and the pooled correlation over all scored tokens. Punctuation is excluded.

5.1 Layer sensitivity

DiffAlign is first evaluated at every Apertus layer on the 453-document fitting partition. Layer selection uses only this partition; the 51-document validation scores below are diagnostics. Both backbones place their strongest raw cross-lingual signal in an early-to-middle band rather than at the final layer.

Backbone	Representation layer	Fitting macro	Validation macro	Engineered-feature layers
Apertus-v1.5-8B	14	0.2759	0.2411	14, 15, 16, 17, 18
Apertus-v1.5-70B	17	0.2855	0.2450	17, 19, 18, 16, 20

For 8B, layer 14 is best on both the fitting and validation partitions. For 70B, fitting selects layer 17; the diagnostic validation maximum occurs at layer 16 with 0.2477, only 0.0027 above layer 17. The fitting-only selection is retained to preserve the experimental boundary.

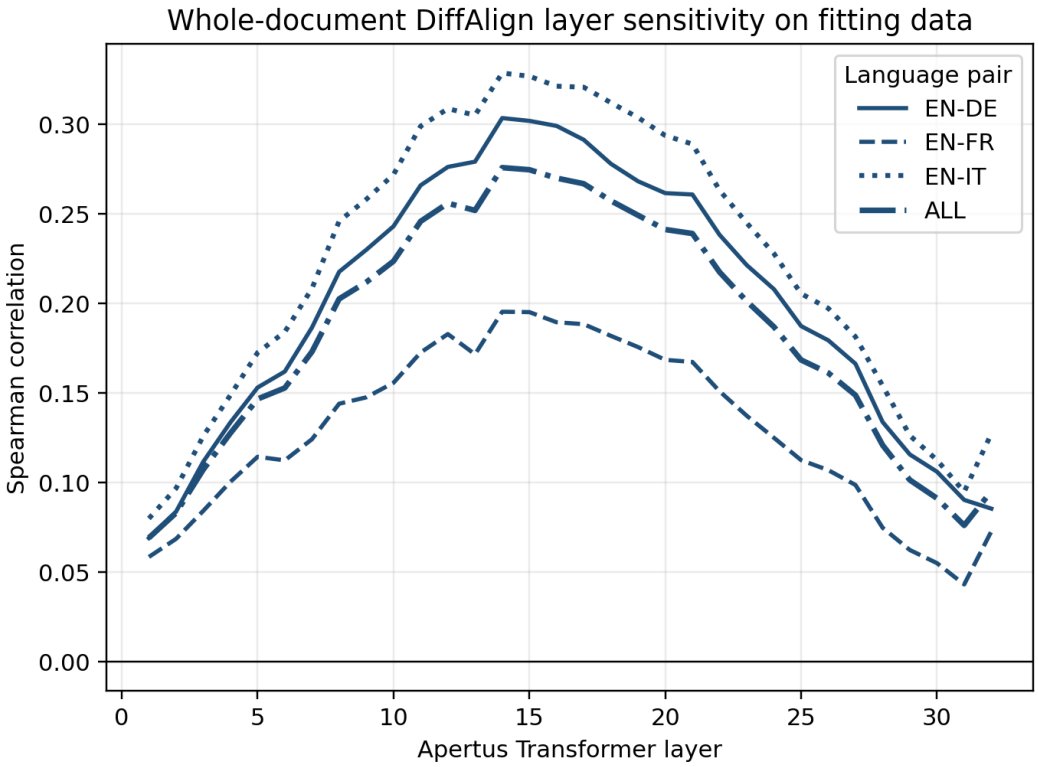


Figure 2. Whole-document DiffAlign Spearman across Apertus-v1.5-8B layers on the fitting partition. Line styles distinguish EN-DE, EN-FR, EN-IT, and ALL; ALL is their unweighted macro average and determines layer selection. The corresponding 70B layer-sensitivity curve is included with the submission.

5.2 8B development diagnostic and frozen test results

Setup	Full-dev macro	Test DE	Test FR	Test IT	Test macro	Change vs. raw	Test pooled
Raw Apertus DiffAlign	0.2730	0.2934	0.1612	0.3350	0.2632	-	0.2672
Representation MLP	**0.4578**	0.3413	0.2177	0.3685	0.3092	+0.0460	0.3199
Engineered gradient boosting	0.3686	0.3678	0.2118	0.3944	0.3247	+0.0614	0.3296
Representation MLP + POS/NER	0.3741	0.3621	**0.2188**	0.3762	0.3190	+0.0558	0.3250
Engineered gradient boosting + POS/NER	0.3715	**0.3705**	0.2158	**0.3961**	**0.3275**	**+0.0643**	**0.3325**

The full-development column is diagnostic: it evaluates all 504 development documents, including the 453 documents used to fit the probes. It is therefore not a held-out generalization estimate. The frozen test columns remain the primary results.

All 168 test documents complete successfully, covering 40,689 German, 46,623 French, and 44,768 Italian tokens (132,080 total). The best system improves raw DiffAlign in every language: +0.0771 for German, +0.0546 for French, and +0.0611 for Italian. Its macro gain is 0.0643 absolute, or 24.4% relative. Engineered features outperform paired raw states in both tagged and untagged comparisons. POS/NER adds 0.0098 to the representation MLP and 0.0028 to the booster on test, although the validation results show that these gains are not uniformly stable. French remains the most difficult direction.

5.3 70B development diagnostic and controlled scaling ablation

The full Apertus-v1.5-70B dependency chain also completes all 168 test documents and 132,080 scored tokens without failure.

70B setup	Full-dev macro	Test DE	Test FR	Test IT	Test macro	Change vs. raw	Test pooled
Raw Apertus DiffAlign	0.2821	0.3048	0.1687	0.3476	0.2737	-	0.2786
Representation MLP	0.3814	0.3588	0.2178	0.3646	0.3137	+0.0400	0.3201
Engineered gradient boosting	0.3778	0.3674	0.1996	0.3804	0.3158	+0.0421	0.3204
Representation MLP + POS/NER	**0.4003**	0.3555	**0.2253**	0.3698	**0.3168**	**+0.0431**	**0.3244**
Engineered gradient boosting + POS/NER	0.3790	**0.3677**	0.2007	**0.3812**	0.3165	+0.0428	0.3213

As in the 8B table, the full-development scores are diagnostic because they include the documents used to fit the probes; the frozen test scores provide the controlled generalization comparison.

Relative to 8B, 70B raises raw DiffAlign by 0.0105 macro and the plain representation MLP by 0.0045. It does not improve either engineered booster or the best final score: the strongest 70B head remains 0.0106 below the 8B primary system. Increased backbone capacity therefore modestly improves raw cross-lingual geometry but does not automatically improve a lightweight probe fitted on the same limited data. No post-hoc 8B/70B ensemble is formed.

5.4 Protocol checks

The frozen result artifacts record test_data_used_for_training_or_selection=false, zero failed documents, and the same token counts for every method. The Docker replay checks the method set, document and token counts, and recomputed macro averages. Predictions on the complete development split are retained only as diagnostics because 453 of its 504 documents are also used to fit the probes; they are not presented as generalization estimates. The exact frozen aggregates are included as final_test_8b_summary.json and final_test_70b_summary.json.

5.5 Prompted baseline audit

During development we attempted the paper-style prompted token-label format. On 50 German development examples, Apertus-8B produced only one structurally valid side out of 100; the rest had label-count or structured-output violations. Its nominal correlation is not treated as a valid baseline. Raw frozen-Apertus DiffAlign is the complete-coverage baseline reported above.

6. Limitations

- Macro Spearman 0.3275 remains well below perfect token ranking.
- French performance is substantially lower than German and Italian.
- Whole-document nearest neighbours can select repeated or generic words and do not enforce monotonic alignment.
- Engineered features may capture benchmark-specific regularities.
- The 51-document validation set serves both early stopping and development comparison, making its scores optimistic; the frozen test result mitigates this but does not provide multi-seed uncertainty.
- POS and NER depend on auxiliary taggers whose errors can propagate.
- The gated Apertus weights cannot be redistributed, and full reproduction requires CSCS-class GPU resources.
- The prompted baseline was structurally invalid and is not directly comparable to the submitted complete-coverage systems.
- The 70B scaling conclusion is based on one frozen training seed for each model size; multi-seed uncertainty remains unmeasured.

7. Reproducibility

Run the frozen result replay:

```
bash
make run
```

Complete Clariden setup, job commands, output paths, model revision, and integrity checks are documented in docs/REPRODUCE.md. Layer scans save train, validation, and all-development Spearman for every layer as CSV plus a PNG plot.

Key controls:

- random seed: 42;
- development partition: 453 train / 51 validation documents;
- test documents successfully processed: 168/168;
- test used for development: false;
- Apertus frozen: true;
- model passes in final test evaluation: 336.

8. Next steps

1. **Improve French robustness.** French remains the weakest language pair, so future experiments should use language-balanced batches, language-specific error analysis, and multiple random seeds to determine whether the gap comes from data composition, representation quality, or probe instability.
2. **Constrain cross-document alignment.** Nearest-neighbour DiffAlign can match repeated or generic words anywhere in the counterpart document. Monotonic, optimal-transport, or document-structure constraints could reduce these false matches while preserving a score for every token.
3. **Strengthen feature learning.** The current POS and NER tags provide only small improvements. A useful extension is to compare them with linguistic attributes predicted directly from Apertus and to learn how evidence from several semantic-sensitive layers should be combined.
4. **Evaluate stronger lightweight heads.** Multi-seed experiments with calibrated ensembles, pairwise ranking losses, and language-aware probes could test whether the remaining performance gap can be reduced without fine-tuning the Apertus backbone.
5. **Establish a valid prompted baseline.** The exploratory prompted system frequently violated the required token-label structure. Constrained decoding or exact schema enforcement would enable a complete-coverage comparison with the representation-based systems.

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